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*An academic department with five faculty members narrowed its choice for department head to either candidate A or candidate B. Each member then voted on a slip of paper for one of the candidates. Suppose there are actually three votes for A and two for B . If the slips are selected for tallying in random order, what is the probability that A remains ahead of B throughout the vote count (e.g., this event occurs if the selected ordering is A A B A B , but not for A B B A A )?*

Number of combinations: 10

Number of combinations with Order: 3

A A A B B

A A B A B

A A B A B

A B A A B

A A B B A

A B A B A

A B A A B

A B A B A

A B B A A

B A A A B

B A A B A

B A B A A

B B A A A

Probability: 3/10